

Incident Triage Agent

Architecture, Working Flow, Prediction Engine & Token Optimization

A detailed technical walkthrough

1. Project Overview

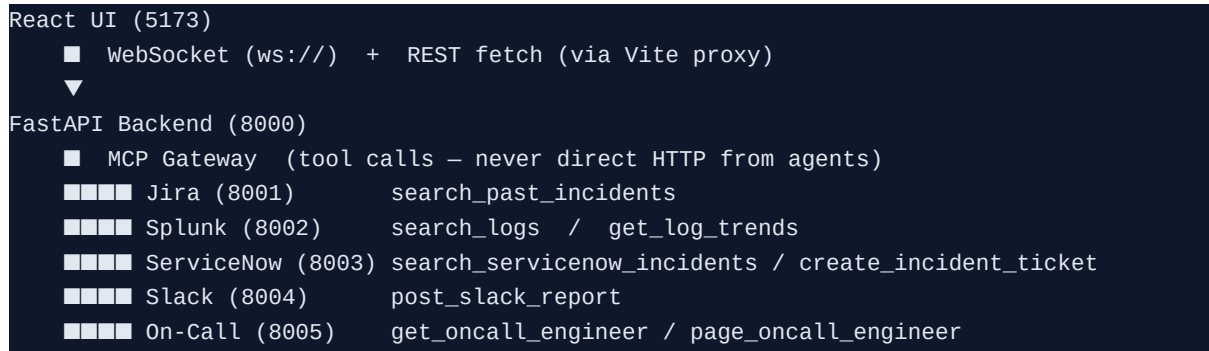
The Incident Triage Agent is an AI-powered system that automatically diagnoses production alerts, identifies root causes, and pages the correct on-call engineer — all without human intervention. It also **predicts incidents before they occur** by continuously monitoring error-rate trends across all services.

The stack: **React + TypeScript** frontend, **FastAPI** backend, **LangGraph** state machine, **Azure OpenAI GPT-4o** for reasoning, **Azure Database for PostgreSQL + pgvector** for runbook retrieval, and five mock microservices (Jira, Splunk, ServiceNow, Slack, On-Call). It is deployed on **Azure Container Apps** (see Section 9).

2. System Architecture

Component	Port	Role
React UI	5173	Live dashboard — alert feed, agent flow panel, triage report
FastAPI Backend	8000	REST API + WebSocket hub + LangGraph pipeline orchestrator
Mock Jira	8001	Historical incident tickets (past + currently open)
Mock Splunk/ELK	8002	Application logs + error-rate trend data for predictions
Mock ServiceNow	8003	ITSM tickets — search, get, create
Mock Slack	8004	Receives and stores formatted incident reports
Mock On-Call	8005	Shift-based engineer routing + page events

Communication topology:



3. Reactive Triage Flow (Alert → Resolution)

Step 1 — Alert Ingestion

An external monitoring system (or the test script) sends a JSON payload to **POST /alert**. The payload contains the service name, alert type, severity, affected endpoints, and current metrics.

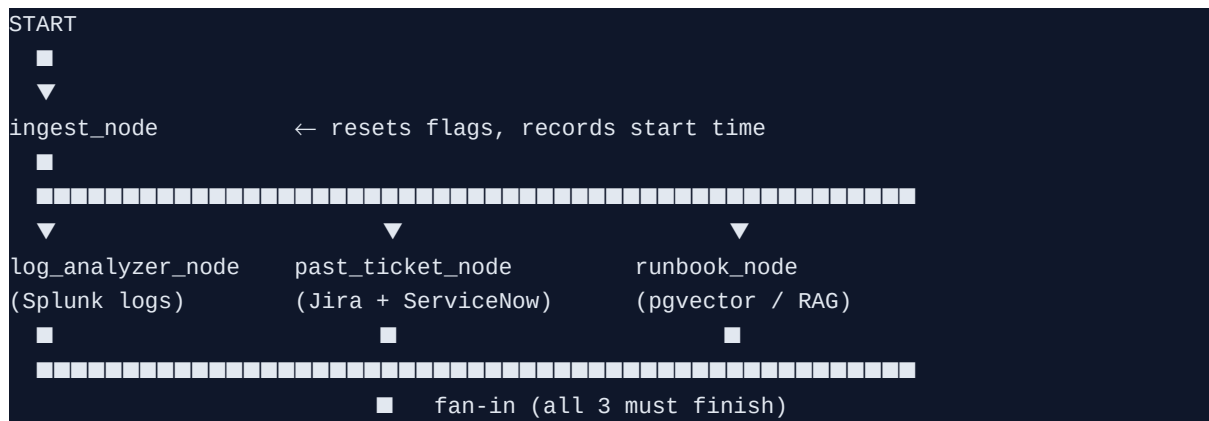
```
python scripts/push_alert.py --scenario db_timeout
■
■■■ POST http://localhost:8000/alert
  {
    "service": "payment-service",
    "alert_type": "DB_CONNECTION_TIMEOUT",
    "severity": "P1",
    "metrics": {"error_rate": "23%", "latency_p99": "8200ms"}
  }
```

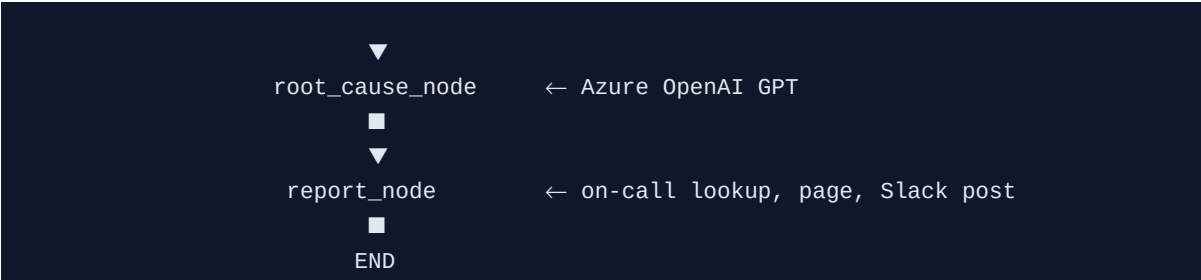
The backend immediately:

- Creates an incident record in memory with status **ingested**.
- Broadcasts a **new_alert** WebSocket event — the UI card appears instantly.
- Returns HTTP 202 Accepted (non-blocking).
- Spawns the triage pipeline as a background asyncio task.

Step 2 — LangGraph State Machine

The triage pipeline is a **directed acyclic graph** built with LangGraph. Each node is an async function that reads from and writes to a shared **IncidentState** dictionary. The graph compiles to a runnable that streams intermediate state updates — every node completion fires a WebSocket broadcast so the UI agent-flow panel updates in real time.





Step 3 — Parallel Evidence Gathering

Three agents run **simultaneously** after ingest_node completes:

Agent	Tool Called	Data Fetched
LogAnalyzerAgent	search_logs → Splunk (8002)	Error + FATAL logs for the service in the last 30 min. Includes scenario-specific anchor logs injected by the mock.
PastTicketAgent	search_past_incidents → Jira (8001) search_servicenow_incidents → ServiceNow (8002)	Historical resolved incidents matching the service + alert type. Also includes currently OPEN / IN-PROGRESS tickets.
RunbookAgent	RAGRetriever.search() → Azure Postgres (pgvector)	Top-5 semantically similar runbook chunks. Embedded with Azure OpenAI and stored in pgvector.

Step 4 — Root Cause Synthesis (Azure OpenAI GPT)

Once all three parallel agents finish, **root_cause_node** calls Azure OpenAI with a structured prompt that bundles all gathered evidence:

```
=== ALERT DETAILS ===          ← original alert payload
=== LOG ANALYSIS FINDINGS === ← Splunk error patterns
=== SIMILAR PAST INCIDENTS === ← resolved Jira + ServiceNow tickets
=== CURRENTLY OPEN TICKETS === ← active in-progress tickets (!)
=== RUNBOOK GUIDANCE ===      ← RAG-retrieved runbook text
```

The LLM (temperature=0) responds with strict JSON:

- **hypotheses** — Top 3 root cause hypotheses ranked by confidence (High/Medium/Low), each with evidence quotes and remediation steps.
- **remediation_checklist** — 10 prioritised actions with owner and time estimate per step.
- **escalation** — Boolean required flag, priority level, team to escalate to, and reason.

Open tickets are critical inputs here — if Jira/ServiceNow already has an active investigation for the same service and error code, the LLM references those ticket IDs in its evidence and remediation steps.

Step 5 — Report Generation, On-Call Paging & Slack

report_node assembles the final report and performs three actions:

Action	How	Condition
On-call lookup	GET /api/oncall/current?service=<name> Mock On-Call (8005)	Always — resolves the team and current shift engineer (morning/afternoon)
Auto-page engineer	POST /api/oncall/page Mock On-Call (8005)	Only if: escalation.required=true AND severity=P1

Post to Slack	POST /api/slack/post Mock Slack (8004)	Always — full structured report sent to the incident channel
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Every state change throughout steps 1–5 is broadcast over WebSocket so the React UI updates live: the agent-flow panel shows each node going from pending → running → completed, and the progress bar advances in real time.

Step 6 — Post-Mortem (human-triggered)

After the incident reaches **done** status, an engineer clicks Resolve in the UI, which calls **POST /incidents/{id}/resolve** with a resolution summary, confirmed root cause, and who resolved it. **PostMortemAgent** runs another Azure OpenAI call to generate a full post-mortem document, stored at **GET /post-mortems/{id}**.

4. Predictive Monitoring (Before the alert fires)

The predictive path runs completely independently of the reactive triage pipeline. It is a background asyncio loop that starts at application startup and polls Splunk trend data for all monitored services on a configurable interval.

4.1 How it starts

```
# backend/main.py – FastAPI lifespan startup
asyncio.create_task(run_predictive_monitor())

async def run_predictive_monitor():
    await asyncio.sleep(15)  # wait for mocks to start
    while True:
        agent = PredictiveMonitorAgent(mcp_gateway=MCPGateway())
        predictions = await agent.run()  # check all 5 services
        for pred in predictions:
            await manager.broadcast({"event": "prediction", ...})
        await asyncio.sleep(interval)  # configured in .env
```

4.2 Per-service trend check

For each of the 5 monitored services, the agent calls **get_log_trends** via the MCP Gateway, which hits **GET /api/logs/trends?service=<name>** on Mock Splunk (8002). The response contains:

Field	Meaning
current_error_rate_pct	Live error rate (%) right now
baseline_error_rate_pct	Normal/expected error rate for this service
trend	"increasing" "stable-elevated" "stable"
rate_of_change_per_min	How fast the error rate is growing (% per minute)
anomaly_score	0.0–1.0 composite anomaly score
predicted_threshold_breach_minutes	ETA until SLA threshold is breached
top_error_codes	e.g. ["CONN_TIMEOUT_5023", "DB_POOL_EXHAUSTED"]

4.3 Prediction decision logic

The agent fires a prediction if **ANY ONE** of three conditions is true:

```
ANOMALY_THRESHOLD      = 0.55    # anomaly_score > 0.55
ERROR_RATE_MULTIPLIER  = 2.5      # current > baseline × 2.5
RATE_OF_CHANGE_THRESHOLD = 0.25   # roc > 0.25 %/min AND trend = "increasing"

def _should_predict(trend):
    return (
        anomaly_score > 0.55
        OR current_error_rate > baseline * 2.5
        OR (trend == "increasing" AND rate_of_change > 0.25)
    )
```

4.4 Confidence and severity mapping

anomaly_score range	Confidence	Predicted Severity
> 0.8	High	P2
0.6–0.8	Medium	P3
< 0.6	Low	P3

4.5 Alert type inference from error codes

The top error codes from the trend response are mapped to known alert types:

```
CONN_TIMEOUT_5023 → DB_CONNECTION_TIMEOUT
DB_POOL_EXHAUSTED → DB_CONNECTION_TIMEOUT
CACHE_MISS_HIGH   → HIGH_ERROR_RATE
OOM_ERROR          → MEMORY_LEAK
HEAP_HIGH          → MEMORY_LEAK
NPE_CHECKOUT       → DEPLOY_REGRESSION
DEPENDENCY_503     → DEPENDENCY_FAILURE
SMS_TIMEOUT        → DEPENDENCY_FAILURE
```

4.6 How the mock generates spikes (demo realism)

The Mock Splunk trends endpoint injects random spikes so predictions fire occasionally but not on every poll cycle. The spike logic:

```
# mock_splunk.py – /api/logs/trends
spike = random.choice([0, 0, 0, random.uniform(2.0, 6.0)])
# 3 out of 4 polls → no spike (stable)
# 1 out of 4 polls → spike of 2x-6x baseline injected

anomaly = (current - baseline) / baseline * 0.5 + random.uniform(0, 0.25)
# anomaly_score is proportional to how far current deviates from baseline
```

4.7 What happens in the UI when a prediction fires

The backend broadcasts a **prediction** WebSocket event. The React UI adds a new  card to the alert feed (visually distinct from real incidents). Selecting it shows a prediction detail view — no agent pipeline runs, because no incident has occurred yet. If a real alert later fires for the same service, it appears as a

separate incident card and the full triage pipeline runs.

5. RAG Pipeline (Runbook Retrieval)

Retrieval-Augmented Generation provides the LLM with relevant runbook knowledge at triage time, grounding its reasoning in actual operational procedures.

Build phase (one-off job — backend/rag/pipeline.py):

```
backend/rag/pipeline.py (run as the triage-index-build ACA Job)
1. Read runbook markdown + past incidents from backend/rag/knowledge_base/
2. Split into overlapping chunks (~400 words each)
3. Call Azure OpenAI Embeddings (text-embedding-3-small) → 1536-dim vector
4. Upsert vectors + metadata into Azure Postgres via langchain-postgres PGVector
5. Collection 'incident_triage_kb' (pre_delete_collection → clean rebuild)
```

Query phase (runs inside runbook_node per incident):

```
RAGRetriever.search(query="DB_CONNECTION_TIMEOUT payment-service")
1. Embed the query string via Azure OpenAI Embeddings
2. PGVector.similarity_search_with_score(query, k=5) ← pgvector cosine search
3. Return top-5 chunks ranked by similarity score
4. Chunks are concatenated and passed as "RUNBOOK GUIDANCE" in the LLM prompt
```

6. MCP Gateway (The Tool Bus)

Every external call — Splunk, Jira, ServiceNow, Slack, On-Call — goes through

MCPGateway.call_tool(name, params). Agents never make HTTP calls directly. This is the **Model Context Protocol** pattern: a registry of named, typed tools with JSON schemas, making it trivial to swap mock services for real ones.

Tool Name	Routes To	Used By
search_logs	Splunk /api/logs/search	LogAnalyzerAgent
get_log_trends	Splunk /api/logs/trends	PredictiveMonitorAgent
search_past_incidents	Jira /api/incidents/search	PastTicketAgent
search_servicenow_incidents	ServiceNow /api/incidents/search	PastTicketAgent
get_incident_details	ServiceNow /api/incidents/{id}	PastTicketAgent
create_incident_ticket	ServiceNow POST /api/incidents	available for future use
post_slack_report	Slack /api/slack/post	report_node
get_oncall_engineer	On-Call /api/oncall/current	report_node
page_oncall_engineer	On-Call /api/oncall/page	report_node (P1 auto-page)

7. WebSocket Real-Time Event Flow

The React UI maintains a persistent WebSocket connection to **ws://localhost:8000/ws** with automatic 3-second reconnection. The backend broadcasts the following event types:

Event	When fired	UI reaction
connected	Client connects	Fetch full incident list
new_alert	POST /alert received	New incident card appears, selected
incident_update	Each pipeline node completes	Progress bar, agent steps update
prediction	PredictiveMonitor fires	■■ prediction card appears
post_mortem_complete	PostMortemAgent finishes	Incident status → resolved

8. System Prompts & Token Optimization

8.1 What is a System Prompt?

A **system prompt** is the static instruction block sent to the LLM before any user content on every call. It defines the model's persona, output contract, task scope, and constraints. It never changes between requests — only the **user message** (HumanMessage) changes with each incident's live data.

```
messages = [  
    SystemMessage(content='You are a principal SRE...') # static every call  
    HumanMessage(content='Alert: DB_TIMEOUT | Logs: ...') # dynamic per incident  
]
```

Because the system prompt is **repeated on every LLM call**, keeping it concise directly reduces cost and latency. Each unnecessary word in a system prompt is billed on every single triage.

8.2 System Prompts in This Project

Every agent has its own system prompt defining a different specialist role:

Agent	Persona defined	Output format
LogAnalyzerAgent	Expert SRE log analyst	JSON — error spikes, exceptions, latency
PastTicketAgent	Incident management specialist	JSON — ranked_incidents + open_tickets
RunbookAgent	Senior SRE runbook specialist	Plain text — 4 markdown sections
RootCauseAgent	Principal SRE / incident commander	JSON — hypotheses + checklist + escalation
PostMortemAgent	Senior SRE post-mortem writer	JSON — full post-mortem document

8.3 Token Optimization — Before vs After

Six optimizations were applied. The table below shows the measured token reduction per agent:

Agent	Sys tokens before	Sys tokens after	Saved	max_tokens before	max_tokens after
LogAnalyzerAgent	186	102	-84	1500	1000
PastTicketAgent	197	114	-83	2500	1500
RunbookAgent	130	54	-76	1500	1000
RootCauseAgent	379	144	-235	3000	2000

Embeddings	Azure OpenAI — text-embedding-3-small (1536-dim)
Vector store	Azure Database for PostgreSQL Flexible Server + pgvector (langchain-postgres)
Agent tooling	MCP Gateway — 9 typed tools fronting 5 mock services
Compute	Azure Container Apps (Consumption plan)
Image registry	Azure Container Registry (Basic)
Secrets / identity	Azure Key Vault (access policies) + user-assigned Managed Identity
Observability	Azure Log Analytics (ACA-integrated, 1 GB/day cap)
CI/CD	GitHub Actions (test → ACR build → containerapp update) — optional
Infrastructure	Bicep — deploy/azure/main.bicep

Azure deployment topology

```
Azure Container Apps environment (region: eastus2)

  triage-frontend  EXTERNAL ingress (nginx :8080) ■■■ public HTTPS URL
    ■ reverse-proxies REST + /ws (HTTP/1.1)
    ▼
  triage-backend   INTERNAL ingress (:8000) · 1 replica, sticky sessions
    ■ (stateful: in-memory state + WebSocket + monitor loop)
    ■■■■ mock-jira / splunk / servicenow / slack / oncall (INTERNAL, scale-to-zero)
    ■■■■ Azure OpenAI (gpt-4o + text-embedding-3-small)
    ■■■■ Azure PostgreSQL (pgvector collection: incident_triage_kb)

  triage-index-build  manual ACA Job → python -m backend.rag.pipeline

Supporting: Azure Container Registry · Key Vault · Managed Identity · Log Analytics
```

Resource sizing (demo / low-usage)

App	Replicas	CPU / Memory	Always-on
triage-backend	1 → 1	0.5 / 1.0 Gi	Yes (stateful)
mock-* (x5)	0 → 1	0.25 / 0.5 Gi	No — scale to zero
triage-frontend	0 → 1	0.25 / 0.5 Gi	No — scale to zero
triage-index-build	job	0.5 / 1.0 Gi	No — on demand

Deploy from a fresh Azure Cloud Shell

```
# 1. set the two secrets (Azure OpenAI key + Postgres URL)
export AZURE_OPENAI_API_KEY=''
export DATABASE_URL='postgresql+psycopg://user:pwd@host:5432/postgres?sslmode=require'

# 2. clone + run the one-shot deploy (idempotent – re-run any time)
git clone https://github.com/Aswin6427/incident-triage-agent.git
cd incident-triage-agent
bash deploy/azure/cloud-shell-deploy.sh
```

```
# It provisions infra (Bicep), builds + pushes images to ACR, rolls them
# onto all apps, runs the index-build job, and prints the live frontend URL.
```

Runs entirely as a Contributor (no role assignments): ACR uses admin credentials and Key Vault uses access policies. CI/CD via GitHub Actions is optional and needs a one-time admin OIDC setup.

10. End-to-End Summary

Path	In one sentence
Reactive triage	Alert in → 3 agents gather evidence in parallel (Splunk + Jira/ServiceNow + RAG) → Azure OpenAI syn
Predictive monitoring	Background loop polls Splunk trend data every N seconds → 3 statistical thresholds detect anomalies →
RAG knowledge base	Runbook documents pre-embedded into Azure Postgres (pgvector) by a one-off job → semantically sea
MCP Gateway	Single tool-call bus between agents and all external systems — 9 registered tools, all swappable for rea